



The sociology of technology before the turn to technology



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ABSTRACT

This project recommences an underdeveloped conversation between the sociology of technology and classical sociology. There was a vibrant and consistent interest in technology among sociology's founders between Marx and Ogburn and revisiting this tradition is beneficial for contemporary sociological studies of technology. In addition to functioning as exemplars of excellence for the sociology of technology, classical sociology provides distinctive and important considerations and contributions, including: the potential benefits of borrowing technology (Veblen), the ecological influences on technological development and use (Cooley), the impact of technology on science (Mauss), and the rationalization of technology (Weber). Most importantly, classical sociology offers partial though unique frameworks for examining technology in society and vice versa, frameworks that are novel precisely because they are out of sync with recent trends.

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1. Introduction

I share the commitment of Weinstein [92] and Westrum [94] to the fruitfulness and contemporary relevance of classical sociological insights for the sociology of technology. However, I problematize and parse out the argument that, “[a]fter Marx [until the 1920s, i.e., Ogburn] there was a long hiatus in social thought about technology. Although there were important social theorists who wrote on the subject, systematic attention to technology and its social relations was absent” [94: p. 50, cf. 92: ch. 2]. A good deal of work within and outside the subfield has shown the depth, complexity, and dynamism of Marx's analysis of technology [e.g., 5,39,45,59: ch. 2, 60: pp. 29–39,93,94: ch. 2], thereby correcting the view of Marx as the archetypal technological determinist. To a lesser extent, William F. Ogburn—if we conceive of his early work as part of the classical period—has received a fair amount of attention in the subfield [e.g., 14,32,92,94: ch. 3].¹ It is correct to assert that sociological analysis of technology tapered between Marx and Ogburn, but there are a number of notable exceptions during and immediately following this period. In addition to reinforcing and deepening Weinstein's [92: p. 46] argument that Thorstein Veblen's system of thought was “a conscious and explicit

sociology of technology,” I show that a series of other classical sociologists offer the subfield important insights: Charles H. Cooley, Marcel Mauss, and Max Weber.²

While acknowledging labor process studies' debt to Marx or citing the Ogburn tradition, some contemporary sociologists consider the sociological study of technology a more recent affair. Woolgar [97] famously announced a “turn to technology” in the

² A few justifications concerning the exclusion and inclusion of particular sociologists and related social scientists seem necessary (exclusion is chiefly due to length restraints). The goal was to select classical sociologists (1) whose ideas I believe contribute to contemporary technology studies and (2) whose contributions primarily sit chronologically in between the work of Marx and Ogburn. The latter were both excluded due to a larger awareness of their ideas among contemporary theorists of technology (see introductory citations). Cooley, although a social psychologist, was included as a classical sociologist because he is often considered one [e.g., 11,34], he worked in a sociology department, and his social psychology is sociological. Early American sociologists of technology after Ogburn, including those associated with the Ogburn tradition [e.g., 25], were excluded because their work largely falls outside what is typically considered the classical period [e.g., 12,13] or was short-lived [e.g., 3]. Spencer's [71: Part 8] theory of technological change in industrial progress, Simmel's [e.g., 65: ch. 6] critique of autonomous technology as objective culture [24], and Scheler's [e.g., 63: ch. 5] critique of “life's” enslavement by the “tool” were excluded due to length restraints and because I have discussed Simmel's and Scheler's sociologies of technology together elsewhere [28]. Additionally, sociological non-sociologists, such as Mumford [54] and Spengler [70], were excluded from analysis due to length. (As one reviewer pointed out, Marx would also be more accurately branded a “sociological non-sociologist” seeing as he associated sociology with Comte's positivism and never aimed to be a sociologist [41: p. 9].)

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¹ See Duncan [16: pp. 352–353] for complete bibliography of Ogburn's works on technology.

social studies of science in 1991, Pinch and Bijker [58: p. 25] stated that “[t]here have been some limited attempts in recent years to launch [a sociology of technology]” before their approach, or, in Russell’s [61: p. 337] reply to the former piece: “technology studies have hardly started.” Certainly, sociological inquiries of technological development and impacts have become more important in the field in recent decades—from microsociology [6: pp. 539f] to environmental sociology [27: pp. 67f]—and it is clear that sociologists, no matter the specialty, should critically reflect on modern technology’s place in their teaching and research [98]. However, perhaps only Weinstein [92] and Westrum [94]—with notable other exceptions discussed in the proceeding sections—take seriously insights from other “old” sociologies of technology during the resurgence in the sociological study of technology since the mid-1970s. Tracing the sociology of technology back to the early modern period, Weinstein [92] showed that modern technology and social science were twin-born, with the early moralists and political economists keen to the moral impact and social potentials of technology. If we are to place modern disciplinary distinctions on this history, Weinstein suggests that an interest in technology from a sociological perspective began with its founder, Comte, who, originated sociology in order to help bring about a Saint-Simonian technocratic order, requiring an integration of technology and sociology. Marx, of course, sustained a sociological analysis of technology, as highlighted above. However, following the institutionalization and specialization of the sciences, the sociology of technology following Marx and preceding Ogburn supposedly became more “passive” and less overt [92,94]. According to Weinstein [92: p. 35], although Weber’s academic sociology contains valuable insights for the sociology of technology, it neglected a systematic analysis of technology, a turn that has influenced the discipline at large. In America too—partially due to the influence of Weberian sociology, a focus on perfecting methodological techniques, and increasing academic specialization—little attention was devoted to sociological analysis of technology, with two exceptions: Ogburn and, earlier, Veblen. However, as Westrum [94: ch. 3] has shown, Ogburn’s near-technological determinism “died out” and the death of Ogburn school—ironically symbolized by the publication of a textbook in 1957 that was intended to begin a new field [2]—almost killed the sociology of technology until its resurgence in the mid-1970s.

The purpose of this project is twofold. First, I hope that the article acts as a stimulus for contemporary sociologists of technology to revisit classical sociological works. None of the classical sociologists receive the in-depth attention that they deserve as the goal is to recommence a conversation with the classics. However, I do provide enough explication and interpretation to point interested sociologists of technology in the right direction for deeper investigation. Nor do I provide any serious critiques of the thinkers covered. If this project is an illustration of Adorno’s [1: p. 4] warning that theoretical perspectives today mirror commodities in a marketplace, where “[e]ach one is offered as a possibility among competing options,” it is due to a conviction that something has been lost without maintaining a conversation with classical thinkers as a whole, as proposed in the concluding comments. The point is to rekindle this conversation, not to immediately limit it. Second, I show that there was a vibrant and consistent interest in technology among sociology’s forefathers. By rethinking prominent classical sociologists as early representatives of the sociology of technology, the goal is for the article to also spark new interest in the sociology of technology among scholars of classical sociology formerly unconcerned with the sociological aspects of technology. After detailing classical sociology’s diverse and unique conceptualizations of technology-society interactions, I conclude with a statement explaining the potential fruitfulness of revisiting the

classics in the sociology of technology.

2. Veblen: sociologist of technology *par excellence*

Technology was the central, though not sole determining [55: p. xxv], variable in Veblen’s sociology. He offers the subfield insights on the social nature of technology, technological development and its relation to social change, and the diffusion and borrowing of technological innovations. Further, his critique of modern capitalism, monopolized and primarily concerned with increasing profits through finance, and normative views regarding an alternative social future are rooted in an analysis of technological change and its social conditions.³ While Veblen’s views on technology have received attention in institutional economics [e.g., 29,85] and the history of technology [35], it is surprising and unfortunate that he is imperceptible in contemporary sociological studies of technology. Perhaps this is due to his relative neglect in studies of classical sociology [67]. Although none of his works can be given adequate treatment here, his catalog provides one of the most systematic and penetrating analyses of technology in the discipline’s history.

Technology, in Veblen [79: p. 103], was interpreted as a strictly social affair:

[t]echnological knowledge is of the nature of a common stock, held and carried forward collectively by the community, which is in this relation to be conceived as a going concern. The state of the industrial arts is a fact of group life, not of individual or private initiative or innovation. It is an affair of the collectivity, not a creative achievement of individuals working self-sufficiently in severalty or in isolation. In the main, the state of the industrial arts is always a heritage out of the past; it is always in process of change, perhaps, but the substantial body of it is knowledge that has come down from earlier generations.

Although individuals contribute to the current state of the industrial arts, any advancement and innovation, even in specialized sectors, comes from their familiarity with the existing “immaterial equipment,” or state of technological knowledge. Innovations from the “savage state of the industrial arts” to the “machine age” are cumulative developments in the “common stock of technology” [79: p. 104]. In Veblen [76: pp. 131ff, 79: ch. 3], social evolution itself is a cumulative and dynamic process of technological development and institutional adaptations, when institutions are understood as “prevalent habits of thought with respect to particular relations and particular functions of the individual and of the community” [76: p. 132]. However, the state of the industrial arts cannot be reduced to the tools and machines utilized. For example, in his vivid discussion of the “machine process” of modern industrial production, Veblen [77: p. 9] clarified that the “scope of the process is larger than the machine”: it stretches across all industries, comprises human bodies, the vast and diverse storehouse of knowledge needed to operate the machinery, a wealth of material inputs, and the standardization of time, communication, and consumption, thereby forming a comprehensive system.

The state of technological development conditions and challenges established habits of thought. In his discussion of the evolution of causal thinking, Veblen [78,79: ch. 5] argued that the development of machine technologies has shaped the modern “matter-of-fact” way modern humans, particularly industrial workers, engineers, and scientists, interpret the world in

³ Due to length restraints, I do not review Veblen’s [79] theory of instincts, where the “instinct of workmanship” “underlies” all technological systems.

mechanical and impersonal ways. Not unlike Husserl [33] and Marcuse [43], though without their critical sting, Veblen argued that science—even when the inquiry is non-purposive, in accordance with the “instinct of idle curiosity”—is always potentially applicable to technological aims due to the influence of the machine process:

the canons of validity under whose guidance he [the scientist] works are those imposed by the modern technology, through habituation to its requirements; and therefore his results are available for the technological purpose. ... [Industry has] become the chief force in shaping men's daily life, and therefore the chief facto in shaping men's habits of thought. [78: p. 598]

Although technological innovations “shake up” social arrangements and make way for new modes of thinking and action, there are never “sharp ontological” ruptures from past technological arrangements [29]. Similar to Ogburn [56, cf. 26], Veblen [76: pp. 132f] argued that outmoded institutions, i.e., commonly shared habits of thought and action, lag behind changes in technological adaptations, and are “never in full accord with the requirements of the present” [cf. 73: pp. 177f]. In fact, vested interests in preserving the status quo hold back progress. Conservative vested interests attempt to preserve outmoded institutional frameworks, a foundational claim that was the basis for his views on technological borrowing and his critique of modern capitalism.

Due to vested interests in maintaining outmoded institutions that constrain the efficient use of new technologies, Veblen famously argued that it is beneficial for societies to borrow, rather than innovate, technologies from other societies in his *Imperial Germany and the Industrial Revolution* [80].⁴ By borrowing English industrial technologies without English institutional residues—namely, the consequences of “pecuniary gain of the business men in control” [80: p. 122], such as wasteful spending on advertising and the inability to update the British railway system due to the profit motive—, Germany was able to use them more efficiently. Although the historical evidence Veblen provided cannot be reviewed here, his seemingly “dubious” yet actually “common-place” premise was:

that a given technological system will have an economic value and a cultural incidence on a community which takes it over ready-made, different from the effects it has already wrought in the community from which it is taken over and in which it has cumulatively grown into maturity in correlation with other concomitant changes in the arts of life. [80: p. 88]

Thereby England paid “the penalty for having been thrown into the lead” in two ways: first, in comparison to what they could accomplish without its outmoded institutions and, second, in comparison to the German advantage acquired through borrowing their technologies [80: p. 132].

The inefficiencies of English industry stressed in Veblen's *Imperial Germany* rests on the same premise as his critique of modern capitalism. Gürkan's [29] concisely summarized Veblen's [77,81–83] argument that private ownership of industry in corporate finance capitalism is antithetical to technological innovation and welfare. This antagonism was conceptualized as a conflict between the outmoded institution of private ownership—born of the handicraft period [81: pp. 25f]—and the potential of increasing welfare through technological progress [76: 154f].

“Depersonalized” corporate owners of the twentieth century were separated from industrial production [77: ch. 3,79: ch. 7,81: ch. 3]. These modern “captains of finance” are qualitatively different than the “captains of industry” of old, in that they do not contribute to technological innovation. The extensive nature of the machine process forces the owners to focus on financial matters and deal with industry from afar, as “absentee owners.” For Veblen, the captains of finance are detrimental to technological innovation because profit maximization is now primarily based on financial speculation and maintaining high prices, not industrial efficiency. In fact, industrial innovation *threatens* the captains of finance due to the risk of overproduction. Thus, for Veblen, private ownership and profit are antithetical to technological progress—and therefore, increased welfare—through production restrictions and speculation.

The paradox between the predatory captains of finance and the “common man” also served as the basis for his provocative technocratic views in *The Engineers and the Price System* [82].⁵ For Veblen [77: ch. 9,78: pp. 608–9], the possibility of a decent future depends on the engineers and industrial workers whose minds were shaped by the matter-of-factness of the machine process overcoming the captains of finance. In any other sociologist, detailed discussions of workers being “disciplined” by “the machine,” the modern “mechanical view of life,” etc. would beckon scorn, whereas cautious anticipation is found in Veblen. His anticipation rested on (1) his sharp distinction between “industry” (synonymous with productivity), which functions to meet the material needs of the community, and “business,” which functions to amass wealth, often on the backs of the industrial [15: pp. 37–38] and (2) the belief that the matter-of-fact worldview of engineers and industrial workers would “disallow” the “unearned income” derived from ownership [81: p. 170]. Yet he was not certain that an “industrial republic” would necessarily be advantageous due to instinctual residues from “savage” society [55: p. xlii]. Dowd [15: 153] has argued that *The Engineers and the Price System* was written in a despaired irony at the impossibility of his “soviet of technicians” due to the victorious captains of finance.

It is clear that technology sits at the heart of Veblen's sociology. He detailed the social nature of technological development and diffusion over a century ago and his theory of social evolution, sociology of knowledge, analysis of borrowing, critique of capitalism, and normative views all give due attention to technology-society relations. In addition to acting as an exemplar classical predecessor of the sociology of technology, perhaps Veblen's most unique and compelling contribution for today is the role his sociology of technology plays in his critique finance capitalism. The argument that late capitalism hinders the potential of forming a society that uses productive technologies to better serve human beings foreshadows later, more well-known accounts [e.g., 43] and is still relevant, especially in light of the increasing power of the finance capital [e.g., 23]. Even if one finds Veblen's despairing technocratic leanings problematic, his normative views on modern technology should be revisited and taken seriously: private corporate ownership is the (perhaps insurmountable) barrier standing in the way of using of the machine as a tool for meeting collective needs.

⁴ For concise summaries, see Coser [11: pp. 273–4], Dowd [15: pp. 87–95], or Tilman [73: pp. 32–36].

⁵ Tilman [73: ch. 6] provides an excellent commentary on, and review of the controversy surrounding, Veblen's seemingly contradictory support for syndicalism, on the one hand, as well as centralized control of industry by experts on the other. See Sadowski [62] for a technocracy taxonomy.

3. Cooley's human ecology of "material communication"

The father of symbolic interactionism is unrecognizable in his abstract on street railways, Cooley's [8: p. 71] first publication, where the "social function" of urban transit is said to reconcile a conflict within the "social organism" between the "industrial necessity" for people to work in dense centers and humanity's need to "not live in dense aggregates." However, as Jacobs [34: pp. 36–37] highlighted, Cooley's dissertation, "The Theory of Transportation" [9], marked the beginning of a transformation in Cooley's thinking from Spencerian functionalism to the proto-symbolic interactionism for which he is known today. The earliest flashes of his insights on communication sit awkwardly alongside residues of Spencerian organismic analogies. It is the latter point of tension, coupled with his acute attention to social and ecological conditioning of technological development, which makes this work on transportation valuable for the sociology of technology.

The conceptual building blocks of Cooley's [9: pp. 39–40] sociological theory of transportation are the three mechanisms that characterize transportation: (1) way ("specialized paths over which resistance to movement is artificially diminished"), (2) vehicle (the carrier of physical things being transported, e.g., wheeled cart), and (3) motive force (the force utilized for movement, e.g., animal power). He argued that these three mechanisms are interdependent, in that the development of one depends on developments in others. Together, transportation is the use of natural forces (as motive force) to move vehicles over natural obstacles via the way. Because the way glides over the earth's surface, the development of the way and the vehicles utilized on it are conditioned by, and developed in consideration to, the particularities of the geographical and ecological characteristics. The implications of this claim take up three chapters of his essay, where he outlined how natural objects and processes act as both obstacles to (e.g., mountains), and facilities for (e.g., dry surfaces), transportation, influencing the use and construction of ways and vehicles during different developments of motive force sources (from human strength to chemical power) [9: chs. 2–4]. To my knowledge, his acute attention to ecological characteristics as conditioners of technological development is certainly unique in the sociology of technology.

The physical-ecological conditions that shape transportation technologies are "relative to social development" [9: pp. 39–40]. Social institutions play a central role in shaping transportation technologies and, in Cooley, no single social institution determines the development of transportation technologies. Each institution uses transportation as a tool to achieve differing needs through the movement of instruments and people, and these needs shape ways and vehicles. For example, economic institutions were "behind" the development of the railroad, "urging it on"; the ways (e.g., going over mountains if faster) and vehicles (e.g., the use of horses instead of other animals) of military institutions valued the principle of speed over all others; and the religious institutional need for the pilgrimage was of central importance in the construction of ways [9: pp. 40, 45, 57f]. In short, transportation is a tool of social institutions that Cooley likened to "material communication."

In a brief though enlightening communication typology, Cooley [9: pp. 40–41] differentiated between two forms of communication, which he considered the "threads" that unite societies: time communication and place communication. Time and place communication forms can take place through psychical or material mechanisms. Psychical time communication (the movement of ideas across time, e.g., preservation of thought through writing) and psychical place communication (the movement of ideas across place, e.g., gesturing, telegraphing) fit the conventional uses of the term "communication." But material time communication (e.g., storage) and material place communication do not. For Cooley [9: p.

40], transportation acts as material place communication, or, the "means of material communication between one place and another." He argued that early developments in psychical and material place communication used some of the same ways and vehicles (e.g., the post used to deliver "light commodities" and "thought") [9: p. 60]. Over time, however, the instruments for material and psychical place communication have uncoupled and specialized accordingly. For example, regarding psychical place communication, "since the introduction of the telegraph it may almost be said that there are no place relations" [9: p. 60]. Fore-shadowing his future intellectual trajectory, Cooley closed his communication typology by dismissing it, arguing that the study of material place communication, i.e., transportation, is a merely "superficial analogy" that can only offer a "skin-deep" picture of the monumental significance and meaning of psychical forms of communication.

It is up for debate whether Cooley's movement away from the study of "material communication" to the exclusive study of "psychical communication" was advantageous to the wider discipline, but it is certain that the insights developed at the beginning of his conversion have applications for the sociology of technology. His attention to the physical-ecological conditioning of technological development is especially notable. The essence of Cooley's [9: p. 39] sociology of technology can be summarized as follows: the "[t]he character of transportation as a whole and in detail, at any particular time and throughout its history, is altogether determined by its inter-relations with physical and social forces and conditions. To understand transportation means simply to analyze these inter-relations." The key message the subfield can learn from Cooley is that social and ecological context matter in the development and use of technology.

4. Mauss' "miracle": technology as a "social thing"

Before exploring Marcel Mauss' contributions to the sociology of technology, contributions recently made available to English speakers [53],⁶ something must be said about the inconsistent place of technology in the work of his uncle, Emile Durkheim. Although both Mauss' [47] gift and Durkheim's [19] totem can be pointed to as precursors to the sociological study of things [45: pp. 152–153], Durkheim's potential insights for the sociology of technology in particular depend on the period of his work in question. Until around the late 1890s [64: p. 8], Durkheim [18: p. 314] specified that technological artifacts are an important variable in sociological analysis: "[s]ocial life, which is thus crystallized, as it were, and fixed on material supports, is by just so much externalized, and acts upon us from without" [cf. 17: pp. 115f]. However, after *Suicide*, there was "radical shift" from his interest in technology and the economy toward the sacred, a change in orientation that Schlanger [64: pp. 5–14] argued resulted from Durkheim's attempt to distance himself from accusations of Marxism. If Durkheim was responsible for the exclusion of "technological" distinctions from *Primitive Classification* [20], Mauss' work marked a "redemption" for technology [64], or, the study of techniques, whose proper place is in the social sciences.

Mauss [48] formulated a brief yet systematic statement on the content of, and potentials for, a sociology of technology, a subfield which merits a "formidable place" in sociology that he predicted would have a "brilliant future." Ahead of his time [e.g., 40: p. 9, 66: pp. 93f], Mauss problematized the commonly understood

⁶ Due to Schlanger's [64] excellent introduction to Mauss' writings on technology, this section will merely highlight major themes of interest for the sociology of technology, excluding insights that may be of more interest to anthropologists.

relationship between science and modern technology, i.e., that technology is applied science. Although science can indeed precede technological developments,

in a thousand cases, techniques pose the questions which science resolves and often creates the facts which science mathematise or schematises after the fact. ... The scientific-technical complex forms a single bloc. For example, the oldest calendars are as much the work of farmers as of religious minds or of astrologers; technique, science and myth are there blended. In the same way, pigeons had been selectively bred before Darwin found the notion of natural selection. [48: p. 53]

For Mauss, the distinction between pure and practical reason blur in concrete studies of his “single bloc.” Although Mauss assigned a central place for the study of technology in sociology and emphasized the conditioning features of technology [50: p. 98], he maintained that society conditions technology (e.g., even when machines manufacture machine-manufacturing machines, this process is still conditioned by labor, nature, and the economy) [52: p. 152]. This indeterminism was grounded in his fundamental claim that technology itself is thoroughly social, a “characteristic feature of communal life”: “[a] practical art has two roots – the invention of the movement or the implement, and the tradition of its use, indeed the use itself – and in both respects it is essentially a social thing” [48: p. 51]. Both roots are social in two senses. First, they are communally created and performed concrete practices and, second, they are symbolic, as collective representations transmitted generationally [49: p. 76, 51: p. 98, 52: p. 149, 64: p. 20].

The social nature of technology can also be understood as a paradox: the simultaneous particularity and generality of techniques. On the one hand, a technique is particular to a given society and makes it distinct from others, even acting as a *sign* for the society in question. Mauss [48: p. 52] asserted that social uniqueness via particular techniques can still be seen in the tools of alike modern societies: “the French and the English ... [have] their different spades and shovels, and this difference requires differences in the mode of their use, and vice versa.” On the other hand, technologies are as generalizable as they are particular; they are “eminently liable to borrowing” [48: p. 52]. For Mauss [48: p. 52], technology’s borrowability, its “extraordinary extrasocial position,” surpasses all other social products, including religion, law, and language, because technology mediates society-nature relations. As a means to both “master nature” and form a “compromise between nature and humankind,” technology, unlike religion or language, takes humankind “out of itself”:

[i]n practical arts, human beings make their limits recede. They advance in nature, at the same time above their proper nature, because they adjust it to nature. ... [H]e creates at once his means of living, things purely human, and his thought inscribed in these things. Here is true practical reason being elaborated. [48: pp. 52, 53]

Herein lies Mauss’ [48: p. 52] basis for elevating the instrument, “that miracle,”⁷ on par with the sacred, both in its importance for sociological study and in “rais[ing] humankind above itself.” As

Schlanger [64: p. 22] stressed, Mauss’ studies on technology, in contrast to the intellectual milieu following the First World War of technology running “out of control” [95], pointed to “an ideal of wholeness and of plenitude ... almost of re-enchantment.” For Mauss [46: p. 47], although the international borrowing of techniques across borders had become a necessity in capitalist societies, leading to a number of conflicts “over state secrets to acquire riches,” he maintained that the spread of technological knowledge and techniques will help bring about an “internationally civilised” humanity. Indeed, some of Mauss’ normative remarks concerning technology are almost technocratic [e.g., 52: pp. 152–153]. Technique, in Mauss [46: pp. 47–48], is the seat of reason, equality, and future happiness and, thus, a critical subject matter for social science.

5. Society precedes technology in Weber

Weber’s treatment of technology has received more attention than the other classical sociologists reviewed thus far, from within the subfield [20,31,42] and from a Weberian social theorist [10: ch. 4]. Technology in Weber signified more than an aggregate of instruments and tools. It is also “ways of seeing and thinking, ways of organizing work and people” [42: p. 73, cf. 84: p. 242], as “part of the realm of spirit” [10: p. 11]. In this light, Maley [42] suggested that Weber’s treatment of purposive rationality, its institutionalization in bureaucracy and capitalism, and the resulting loss of meaning [e.g., 89: pp. 139f] is, at bottom, an implicit critique of technology that can be classified alongside positions such as Ellul’s [21, cf. arguments in 22: p. 4, 69: p. 142]. Collins [10] too has highlighted the uniqueness of Weber’s sociology of technology, though emphasizes his insistence that technology is embedded in, and dependent on, the social world [cf. 31]. Similarly, for Hård [30], Weber provided a framework for examining the emergence of technology through a conflict theoretical perspective.

When examining a passage from one of Weber’s not-so-value-free discussions of bureaucratization, it is easy to see why Weber is a critic of technology in Maley and an anti-technological determinist in Collins and Hård:

[a]n inanimate machine is mind objectified. Only this provides it with the power to force men into its service and to dominate their everyday working life as completely as is actually the case in the factory. Objectified intelligence is also that animated machine, the bureaucratic organization ... Together with the inanimate machine it is busy fabricating the shell of bondage which men will perhaps be forced to inhabit some day, as powerless as the fellahs of ancient Egypt. [90: p. 1402, cf. p. 223]

Similarly, Weber’s attention to, and concern with, technology is found in the middle of the most famous paragraph from *The Protestant Ethic* [86: p. 181] on the “iron cage” of bureaucratic-capitalist society “now bound to the technical and economic conditions of machine production” (for model of technological innovation partially inspired by Weber’s sociology of religion, see Coccia [7]). Indeed, Weber [86: pp. 181–182] stated that it is precisely because capitalism “rests on mechanical foundations” that its existence no longer requires religious asceticism’s justification for existence. However, Weber should not be interpreted as a technological determinist; far from it [31: p. 59, 10: p. 25]. In his reply to Sombart’s 1910 lecture “Technology and culture” (which remains untranslated), Weber [87: pp. 27, 31] rejected all forms of determinism, including its technological form, arguing that “the same technology does not always denote the same economy, nor is the reverse always the case. ... If we lay the causal chain before us, it will always go quickly from

⁷ In Mauss [51: 100], a tool is a simple, single piece of matter (e.g., a lever), an instrument is a “combination of tools” (an axe), a machine is “a combination of instruments” (e.g., a bow), and an industry/craft is a “combination of machines” used together to achieve an end (e.g., the boat and tackle in fishing). Another unique feature of Mauss’ [50] work was his insight that not all techniques require tools, instruments, or machines, which he called “techniques of the body” (e.g., swimming).

technological to economic and political, and then from political to religious and then to economic etc.” [cf. 40: p. 6]. This does not mean he rejected technology as a potential “independent variable,” speculating that modern city infrastructure played an important role in the development of modern painting and that there are purely technical considerations in architectural development [87: pp. 29–30]. Yet even when discussing the technological prerequisites for bureaucratization, such as developments in communication infrastructure, he qualified the discussion, clarifying that technological development “alone is not decisive” [90: p. 973]. His anti-determinism, helpful for explaining the “social shaping” of technology [40], can be seen in his discussions of economic, military, and music technologies.

Weber’s [90: pp. 65–67] anti-determinism is analytically founded on his action typologies, where he argued that “technical action” is concerned with the means used to attain an end, without concerning itself with the rationality, quality, or meaning of ends. He was quick to point out that economic action often determines the ends of *Technik*. Despite non-economic influences on technological development, “[t]he fact that what is called the technological development of modern times has been so largely oriented economically to profit-making is one of the fundamental facts of the history of technology” [90: p. 67]. While he did not provide an economic-sociological history of technological development in his *opus* [90: p. 121], he did provide more specific instances of this claim in his *General Economic History* [91: pp. 129, 231, ch. 27], where the proliferation of the “rationalized technologies” of industrial production are preconditioned by predictable, large-scale inputs, a mass market for goods, and free labor and driven by the need to reduce production costs. Collins [10: pp. 23f, 40–48, ch. 4] has already superbly detailed and developed Weber’s conceptualization of industrial technologies as a “dependent variable” elsewhere.

Along with the sociological foundations of industrial technologies, Weber provided a sociological theory of military technologies, specifically how they interact with the “conflict” between charismatic-individual warfare and disciplined warfare. Surprisingly, Weber [90: p. 1150] opened his discussion with a deterministic argument: that the difference between charismatic and disciplined warfare “is to some extent purely determined by technology.” For example, the horse as a military technology preceded the individual-charismatic hero on his chariot. Yet the rest of the section was devoted to showing that the opening claim is backwards, providing a number of examples of developments in military technologies preceded by disciplined warfare.

It is evident that the kind of weapon has been the result and not the cause of discipline. ... Gun powder and all the war techniques associated with it became significant only with the existence of discipline—and to the full extent only with the use of war machinery, which presupposes discipline. [90: pp. 1151, 1152]

Despite the sociological origins of modern military technologies in discipline, these developments have increased the ability to wield technological power in bureaucratized militaries. Just like in the modern factory [91: ch. 27], bureaucratic military organizations are marked by the “concentration of material means of management in the hands of the master” [90: p. 980]. Disciplined warfare has certainly won in the “conflict” between charismatic warfare in the modern “mechanized” military. In modern disciplined warfare, in which officers are “technicians,” there is a “war of machines, and this makes centralized provisioning technically necessary, just as the dominance of the machine in industry promotes the concentration of the means of production and management” [90: p. 981].

Weber’s [88] unfinished sociological study of music provides further insights into the social origins of technological development, specifically of the rationalization of musical instruments in the West. The origins of the standardization of musical instruments occurred in the West due to a constellation of social factors during the thirteenth century, especially the formation of guilds devoted to manufacturing stringed instruments and, later, “the gradual acceptance of instrumentalists beside the singers in the bands of hierarchy, of princes and of communities,” which increased demand for the guilds’ standardized instruments [88: p. 107, cf. 36: pp. 169ff, 74: pp. 637f]. As Käsler [36: pp. 170f] stressed, Weber [88: pp. 117–24] integrated economic, technological, climatic, and cultural variables to explain the development and diffusion of the piano. Due to its bourgeois, indoor home culture, Saxony not only became the center of piano improvements and production, but the piano itself became “a significant piece of middle-class furniture” in northern Europe: “it is no accident that the representatives of pianistic culture are Nordic peoples, climatically house-bound and home-centered in contrast to the South. In southern Europe the cultivation of middle-class home comforts was restricted by climatic and historical factors” [88: p. 120]. For Weber [88: 121–22], the “victory” of the hammer piano over the clavichord and harpsichord resulted not from technical improvements, but from the fame of Mozart and the demand of “music publishers and of concert managers to satisfy the large music consumption of the mass market.” Later, the mechanized mass production of the iron piano in England and America allowed for the piano to diffuse where climatic factors had once restricted its use (i.e., more tropical areas). Here, in agreement with Cooley, social and physical-ecological factors are critical variables in understanding the development and diffusion of technology.

Although Weinstein [92] was correct that the analysis of technology lessened in Weber’s sociology compared to the likes of Marx, his work made an important contribution to the sociology of technology. His attention to the social conditioners of technological development is his most important contribution. Weber showed how economic-productive, military, and even musical technologies are conditioned by preexisting social processes. His anti-determinism will certainly find supporters among contemporary sociologists of technology and shows that the classical period cannot be swept aside as suffering from technological determinism. In addition, sociologists should revisit Weber’s sophisticated early critique of modern technology, as Maley [42] has emphasized. Ironically, Weber’s value-free sociology ought to inspire the sociology of technology, a subfield currently marked by excessive value-neutrality, to make normative commitments.

6. Conclusion: why sociologists of technology should revisit the classics

Technology remained an important concern for sociologists between Marx and Ogburn. Veblen, Cooley, Mauss, and Weber all provide the sociology of technology with insights to think about and frameworks for thinking within (see note 2 for additional classical sociological takes on technology not explored in this analysis). It is unfeasible and unnecessary to provide a synthetic sketch of a single classical sociology of technology because the views summarized above vary widely in focus, purpose, method, and judgement. Rather than formulating an integrative system or merely providing a history of social thought, the purpose of this project is to incite interest in classical sociological theory among contemporary sociologists of technology. The assumption underlying this goal is that pulling old ideas to the surface opens

new doors for exploration. Yet scholars that revisit and reinterpret the ideas put forth in earlier sociological works must inevitably question the importance of such excavation projects. Some believe it is a waste of time to repeatedly analyze the ins and outs of theoretical systems—categorizing and nuancing here, problematizing and revising there—because one “[risks] wandering into the houses of mirrors, brilliant but exitless,” at the expense of pursuing studies of substantive, real life problems [68: p. 10, cf. 75]. Others, however, have provided justifications for “wandering into houses of mirrors” (see Levine [38] for review and typology).

Levine [38] has summarized a number of cases for revisiting the classics that emerge from different narratives of sociology's development. Three are especially applicable for this project. (1) The “humanist” narrative asks us to revisit sociological classics because they are enduring products of genius during a Golden Age of sociology. This is similar to Stinchcombe's [72] notion of classics as “touchstones” or exemplars of excellence. (2) The “positivist” narrative looks to the classics not only as “monuments” of progress toward a “truly scientific” field, but as sources of underexplored concepts, explanations, and data, or what Stinchcombe [72] called “underexploited normal science.” (3) The “pluralist” narrative invites us to revisit the classics on the grounds that they formulated varying partial frameworks and healthy and necessary divisions in thought that can provide “alternative approaches in the face of current orthodoxies” [38: p. 312]. These three narrative-based justifications provide good reasons for sociologists of technology to revisit the classics.

One function of classical sociological works in the humanist narrative is to serve as exemplars of excellence that inspire excellence; as “touchstones” [72]. Citing Levi-Strauss' claim that he would always read passages from Marx's *18th Brumaire* before writing, what Stinchcombe [72] had in mind is the setting of “aesthetic standards” that cannot be dictated by any philosophy of science. For example, Latour has set the aesthetic standard for many contemporary scholars in science and technology studies (playful seriousness, attention to thick descriptions of the mundane, pleasure in subverting common assumptions about how science works, etc.). This is one fruitful reason that contemporary sociologists of technology should revisit the classics. Weber's study of the collective affinities that led to the development and diffusion of the piano, Veblen's institutional analysis of the intricacies of the machine process, or Mauss' programmatic outline for a sociology of technology are exemplars of excellence that may inspire excellence.

The substantive and more important reason sociologists of technology should revisit classical sociological works is that the latter serve as “underexploited normal science” [72: p. 8], what Levine [38] called the “positivist” case for revisiting the classics. More than a depot of hypotheses waiting to be tested, however, this function means that classical works are simply good for thinking: “puzzles one can find in classic works often are more interesting than the puzzle of entering another variable in a model of status attainment” [72: p. 9]. Each thinker discussed above offers the subfield distinctive considerations, e.g., the potential benefits of borrowing technologies from other groups (Veblen), the physical-ecological influences on technological development and use (Cooley), understanding that technology influences science (Mauss), and the rationalization of technology (Weber). Pinch [57] has shown how much traction one gets out of revisiting Goffman, who, though not a classical sociologist, has a “hidden sociology of technology” (e.g., the merry-go-round's role in role distance) that helps us study more mundane technologies.

Perhaps the best reason for sociologists of technology to revisit the classics is the “pluralist” narrative: they offer contemporary sociologists of technology partial though unique lenses for seeing

technology in society and vice versa as well as alternatives to current predominant frameworks. These lenses are novel precisely because they are dated, or, out of sync with recent trends, namely social constructionism and material semiotics (including actor-network theory) [44]. Here, I am in agreement with Winner's [96] assessment of what has been “given up” when studies of technology cast off thinkers like Ellul, Marx, Mumford, and Heidegger in favor of a (still) dominant and one-sided social constructionism [e.g., 4], including a lost understanding of the social impacts of technology, a shortage of structural analyses, and a lack of value commitments. His claims are equally valid when considering the sociology of technology in light of classical thinkers. Our lost understanding of the social impacts of technology—which, as Winner [96: p. 368] rightly reminded us, were never fully understood—may be rejuvenated, for example, by revisiting Veblen's account of how technology use conditions worldviews or Cooley's claim that new technologies can uncouple and specialize “material” and “psychical” communication. The forgotten role of structural (i.e., bigger-than-actant) influences on technological change [cf. 37] can be taken up again through, for example, Veblen's institutionalism or Weber's rationalization thesis. The lack of normative and political stances concerning the “basic commitments and projects of modern technological society” [96: p. 375] could be reevaluated, for example, through Veblen's or Mauss' technocratic leanings or the more gloomy assessment of Weber.

Undoubtedly, we will find the classics lacking in many respects, and this work does not offer a critique. However, forgotten frameworks are all the more fruitful for illuminating new objects for study or recasting commonly studied areas. Contemporary sociologists of technology should revisit the classics because fresh ideas are often the old ones, or, sometimes “staying behind coincides with being ahead” [99: 4].

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